

JOINT DISTRIBUTION-INFORMED SHAPLEY VALUES FOR SPARSE COUNTERFACTUAL EXPLANATIONS

Anonymous authors
Anonymous submission

ABSTRACT

Counterfactual explanations (CE) are intended to make explicit the input perturbations through which a model prediction can be redirected, yet many existing procedures introduce a superfluous number of feature modifications, thereby diminishing both interpretive parsimony and practical actionability. We introduce *COLA*, a model- and generator-agnostic post-hoc framework that refines any supplied CE by estimating, through optimal transport (OT), a coupling between factual and counterfactual sets and by using that coupling to parameterize a Shapley-based attribution (*p*-*SHAP*) that selects a minimal collection of edits while retaining the target effect. On the theoretical side, OT minimizes an upper bound on the W_1 divergence between factual and counterfactual outcomes, and, under mild conditions, the refined counterfactuals are guaranteed not to be more distant from the factual observations than the original counterfactuals. Empirically, across four datasets, twelve models, and five CE generators, COLA attains the same target effects while requiring only 26–45% of the original feature edits. In a small-scale benchmark, COLA further exhibits near-optimal behavior.

The experiment code and software will be released upon acceptance.

1 BACKGROUND

Explainable Artificial Intelligence (XAI) has become indispensable for rendering artificial intelligence systems intelligible, auditable, and worthy of trust (Arrieta et al., 2020; Das & Rad, 2020). Within this broad agenda, feature attributions (FA) methods, including Shapley values (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017), quantify the contribution of each input feature to a machine learning (ML) model’s output, thereby compressing otherwise opaque decision mechanisms into a ranked account of influential variables. In healthcare, for example, Shapley values may foreground factors such as age and medical history, enabling clinicians to interrogate model decisions in relation to clinically meaningful covariates (Ter-Minassian et al., 2023; Nohara et al., 2022). By contrast, counterfactual explanations (CE) (Wachter et al., 2017; Guidotti, 2022) explain predictions by specifying how alterations of input features would induce different outcomes. Although hundreds of CE algorithms have now been proposed (Guidotti, 2022; Verma et al., 2020), a universally adequate method is implausible because existing algorithms are deliberately specialized to distinct scenarios, objectives, and tasks: some seek a single counterfactual instance for each factual observation, others process multiple factual observations jointly and return one or more counterfactuals per observation, and still others operate at the dataset level by identifying global CE directions along which factual instances should move to reach a desired model output (Rawal & Lakkaraju, 2020; Ley et al., 2022; 2023; Carrizosa et al., 2024). In distributional formulations, moreover, the factual observations are treated as an empirical distribution, and the goal is to obtain a counterfactual distribution that preserves distributional shape and comparable costs (You et al., 2025); orthogonally, some algorithms require differentiable models, whereas others are expressly constructed for tree-based or ensemble architectures.

Problem Description and Challenges Building on the substantial methodological foundations surveyed in Appendix A (see Appendix A), we study a deliberately general problem that arises precisely because FA and CE, while both central to interpretability and accountability, are individually incomplete. Used in isolation, FA may identify influential variables without providing implementable courses of action, whereas CE methods may prescribe concrete changes that include redundant or

impractical feature modifications. The resulting question, which motivates the formulation below, is therefore how to retain the actionable orientation of counterfactual reasoning while imposing the parsimony typically associated with feature attribution:

Given a (group of) factual instance(s), how can we devise an action plan that requires the least feature modifications to achieve a desired counterfactual outcome?

$\mathbf{x}$	$\mathbf{z}'$	$\mathbf{z}''$	$\mathbf{y}^*$
$\textcircled{a}$  $\textcircled{R}$	$\textcircled{a}$  $\textcircled{R}$	$\textcircled{a}$  $\textcircled{R}$	$\textcircled{R}$
200 5 No	250 8 Yes	200 5 No	Yes
150 3 No	150 3 No	150 7 Yes	Yes
100 2 No	350 9 Yes	100 2 No	Yes
150 6 No	150 6 No	350 6 Yes	Yes

Figure 1: [Example: User engagement on an e-commerce platform] An e-commerce platform seeks to increase user registrations using observed interaction data, including spending ($\textcircled{a}$), click counts (), and registration status ($\textcircled{R}$). In the factual data ($\mathbf{x}$), no user is registered; action plans $\mathbf{z}'$ and $\mathbf{z}''$ therefore alter user characteristics so as to approach the desired output ($\mathbf{y}^*$) of full registration. Although both plans realize a half counterfactual effect, $\mathbf{z}''$ does so with fewer feature modifications than $\mathbf{z}'$.

Three interrelated challenges make this problem nontrivial. First, because counterfactual objectives are intrinsically task-dependent, it is unrealistic to require a single CE generator to satisfy all use cases. Second, any refinement procedure should avoid restrictive assumptions about the predictive model, such as differentiability or specialized architectural structure, if it is to remain broadly applicable. Third, feature-importance scores can be misleading when they are not coherently tied to the feature changes that actually produce the desired counterfactual effect; equivalently, it is ineffective to perform FA independently of CE to select the most important features to change. As we show later in Result II of Section 6, this decoupling may induce counterproductive feature modifications, or stakeholder actions, because variables that appear globally important need not coincide with the particular pathways through which the intended counterfactual outcome is attained.

Main Contributions We introduce COUNTERFACTUALS WITH LIMITED ACTIONS (COLA), a general post-hoc framework that refines CE across heterogeneous models and CE generators by using an optimal transport (OT)-induced coupling between factual and counterfactual sets as the distributional substrate for Shapley attribution. This construction yields p -SHAP, which subsumes other commonly used Shapley methods under appropriate couplings while providing a modular interface for attribution-guided edit selection. We establish theoretical guarantees showing that OT minimizes an upper bound on the W_1 divergence and that, under mild conditions, the refined counterfactuals are provably no farther from the factual observations than the original counterfactuals. Empirically, across four datasets, twelve models, and five CE methods, COLA with p -SHAP preserves the same target effects while using only 26–45% of the original feature edits; in a small-scale benchmark, it is additionally near-optimal.

To our best knowledge, this is the first method proposed for systematically addressing the problem in Figure 1 *without specifying certain CE algorithms and models*.

2 PROBLEM FORMULATION

We now formalize the problem illustrated in Figure 1. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ denote a black-box ML model, let $\mathbf{x}$ denote observed factual data with n rows and d columns ($\mathbf{x} \in \mathbb{R}^{n \times d}$, $n \geq 1$, and $d \geq 1$), and let $\mathbf{y}^*$ denote the target model output ($\mathbf{y}^* \in \mathbb{R}^m$, $m \geq 1$). The optimization seeks a group of counterfactual data instances $\mathbf{z}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$) that, while respecting a maximum budget C on admissible feature changes, induces model outputs as close as possible to $\mathbf{y}^*$. With D denoting a generic divergence function that measures dissimilarity between two entities, the problem is

formulated as follows.

$$\min_{\mathbf{c}, \mathbf{z}} D(f(\mathbf{z}), \mathbf{y}^*), \quad (1a)$$

$$\text{s.t. } D(\mathbf{z}, \mathbf{x}) \leq \epsilon, \quad (1b)$$

$$\sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (1c)$$

$$z_{ik} \leq x_{ik}(1 - c_{ik}) + Mc_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d, \quad (1d)$$

$$z_{ik} \geq x_{ik}(1 - c_{ik}) - Mc_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (1e)$$

The objective equation (1a) together with the proximity constraint equation (1b) instantiates the standard CE desideratum: $\mathbf{z}$ should make $f(\mathbf{z})$ close to $\mathbf{y}^*$ while remaining sufficiently close to $\mathbf{x}$, so that $\mathbf{z}$ can serve as a contrastive reference explaining why $f(\mathbf{x})$ fails to attain $\mathbf{y}^*$. We deliberately leave D unspecified, thereby allowing divergences such as Euclidean distance, OT, maximum mean discrepancy (MMD), or differences between mean or median model outcomes. Constraints equation (1c)–equation (1e) add the action-budget component to the counterfactual optimization: in addition to optimizing the counterfactual data $\mathbf{z}$, we optimize an indicator matrix $\mathbf{c}$ such that z_{ik} may differ from x_{ik} only when $c_{ik} = 1$. Constraint equation (1c) permits at most C such changes, while equation (1d) and equation (1e) enforce equality $z_{ik} = x_{ik}$ whenever $c_{ik} = 0$ and, when $c_{ik} = 1$, use a sufficiently large constant M to allow the counterfactual entry to vary.

To solve equation (1), we use FA to identify the features whose modification is most consequential and thereby determine the indicator matrix $\mathbf{c}$. The next section reviews standard Shapley-value methods for FA; these methods, together with the proposed variant introduced subsequently, are incorporated into the COLA framework to construct the refined counterfactual $\mathbf{z}$. The resulting optimization remains computationally difficult even for linear models with $d = 1$, as shown in Appendix B.

3 PRELIMINARIES ON SHAPLEY VALUE

This section reviews widely used Shapley-value constructions for FA, since these baselines and our proposed variant are all embedded within the algorithmic framework of COLA. We begin with the cooperative-game-theoretic definition of the Shapley value and then specialize it to several attribution methods commonly used in FA; for a broader taxonomy of Shapley-value methods, see (Sundararajan & Najmi, 2020).

In cooperative game theory, a coalitional game is specified by a finite player set, here interpreted as the feature set $\mathcal{F} = \{1, 2, \dots, d\}$, and by a characteristic function v . The mapping $v : 2^{\mathcal{F}} \rightarrow \mathbb{R}$ assigns to each coalition $S \subseteq \mathcal{F}$ a real-valued payoff $v(S)$, understood as the payoff attainable through cooperation among the members of S , with $v(\emptyset) = 0$. The Shapley value derived from v provides an axiomatic allocation of the grand-coalition payoff $v(\mathcal{F})$ across individual players, and, in the attribution setting considered here, across features.

Before defining the Shapley value, let $\Delta(k, S)$ denote the marginal contribution of player, or feature, k to coalition S ; this quantity is the additional payoff generated when k is incorporated into the coalition:

$$\Delta(k, S) = v(S \cup k) - v(S). \quad (2)$$

The Shapley value ϕ_i for a player i in a cooperative game is then defined in equation (3).

$$\phi_k = \frac{1}{d} \sum_{S \subset \mathcal{F} \setminus \{k\}} \binom{d-1}{|S|}^{-1} \Delta(k, S). \quad (3)$$

Thus ϕ_k averages the normalized marginal contributions $\Delta(k, S)$ of feature k over every subset S that excludes k . The inverse binomial coefficient $\binom{d-1}{|S|}^{-1}$, with d denoting the total number of features, normalizes coalitions of different sizes; after aggregation over all such coalitions and division by d , the expression yields an axiomatic allocation of payoff that represents feature k ’s contribution relative to the full feature collection.

For an arbitrary data point $\mathbf{x}_i$, we write $\mathbf{x}_{i,S}$ for the restriction to features in S and $\mathbf{x}_{i,\mathcal{F} \setminus S}$ for the complementary components, applying the same notational convention to other vectors. We next instantiate this game-theoretic template in several Shapley-value methods used for feature attribution.

Baseline Shapley (B-SHAP). This approach uses a specific baseline data point $\mathbf{r}_j$ as the reference against which any $\mathbf{x}_i$ is evaluated, with the set function defined as

$$v_B^{(i)}(S) = f(\mathbf{x}_{i,S}; \mathbf{r}_{j, \mathcal{F} \setminus S}) - f(\mathbf{r}_j), \quad (4)$$

where feature absence is represented by substituting the corresponding value from the reference baseline data point $\mathbf{r}$ (Lundberg & Lee, 2017; Sun & Sundararajan, 2011; Merrick & Taly, 2020). Consequently, B-SHAP presupposes an exact alignment between $\mathbf{x}$ and $\mathbf{r}$.

Random Baseline Shapley (RB-SHAP). This variant of B-SHAP, used implicitly in (Lundberg & Lee, 2017) and in the widely used ‘‘SHAP’’ library, replaces the fixed baseline with a background distribution. Its characteristic function is

$$v_{RB}^{(i)}(S) = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}_{i,S}; \mathbf{x}'_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}')]. \quad (5)$$

where $\mathcal{D}$ denotes a background distribution, typically instantiated by the training dataset (Lundberg & Lee, 2017; Merrick & Taly, 2020).

Counterfactual Shapley (CF-SHAP). CF-SHAP extends RB-SHAP by taking $\mathcal{D}$ to be a counterfactual distribution defined separately for each factual instance $\mathbf{x}_i$ ($i = 1, 2, \dots, n$).

$$v_{CF}^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{r})]. \quad (6)$$

This construction, adopted in (Albini et al., 2022; Kommiya Mothilal et al., 2021), has been shown to be advantageous for contrastive FA; unlike B-SHAP, it assumes a probabilistic rather than exact alignment between $\mathbf{x}$ and $\mathbf{r}$.

4 THE PROPOSED JOINT-PROBABILITY-INFORMED SHAPLEY (p -SHAP) AND ITS THEORETICAL ASPECTS

(Proposed) p -SHAP We generalize equation (4)–equation (6) by introducing an algorithm A_{Prob} that explicitly returns a joint probability law linking factual and counterfactual observations.¹ Under this probabilistic alignment, p -SHAP is defined as follows:

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r}|\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r})} [f(\mathbf{r})], \quad (7a)$$

$$\text{s.t. } p = A_{\text{Prob}}(\mathbf{x}, \mathbf{r}). \quad (7b)$$

Crucially, in contrast to prior approaches that rely on random baselines, p -SHAP addresses the alignment problem by identifying an optimal coupling through OT. Feature attribution is thereby reframed not merely as a decomposition of a prediction, but as a transport-informed explanation problem in which the cost of moving from factual to counterfactual support is explicitly minimized.

It is important that A_{Prob} is independent of the CE algorithm, but only depends on the generated counterfactual $\mathbf{r}$. By conditioning attribution on these fixed objects rather than on the idiosyncrasies of the generator that produced them, p -SHAP yields a more stable attribution mechanism than CF-SHAP. Contrary to a natural expectation that the generator-defined counterfactual distribution should be preferable, we later demonstrate that OT can be more effective than relying on a counterfactual distribution defined by a CE generation mechanism as done by Albini et al. (2022), in Result II of Section 6 later.

A central instantiation in this paper is the OT problem, equivalently the regularized 2-Wasserstein divergence, defined below. The transportation plan $\mathbf{p}_{\text{OT}}$ obtained by solving this problem is then used as the joint distribution of $\mathbf{x}$ and $\mathbf{r}$ within p -SHAP.

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\mu, \nu)} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 + \varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log \left(\frac{p_{ij}}{\mu_i \nu_j} \right). \quad (8)$$

¹First, p -SHAP reduces to B-SHAP in equation (4) when A_{Prob} defines a joint distribution between $\mathbf{x}$ and $\mathbf{r}$ that indicates an $i \leftrightarrow j$ alignment for a given $\mathbf{x}_i$ and $\mathbf{r}_j$. Second, p -SHAP reduces to RB-SHAP in equation (5) when A_{Prob} is independent of CE and instead associated with an arbitrary distribution $\mathcal{D}$. Third, p -SHAP reduces to CF-SHAP in equation (6) when A_{Prob} is constructed from a known CE distribution.

Here μ and ν denote the marginal distributions of $\mathbf{x}$ and $\mathbf{r}$, respectively, while $\Pi(\mu, \nu)$ denotes the set of admissible joint distributions, or transport plans. The term $\varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log(p_{ij}/(\mu_i \nu_j))$ is the entropic regularizer, with $\varepsilon \geq 0$ as its coefficient; when $\varepsilon > 0$, this regularization improves the computational tractability of OT.

Theoretical Aspects of p -SHAP Because OT minimizes the aggregate feature-modification cost incurred when factual observations $\mathbf{x}$ are transported toward counterfactual observations $\mathbf{r}$, its induced alignment is directly consonant with our objective of attaining counterfactual outcomes through low-cost modifications. Equivalently, it formalizes the cost of modifying $\mathbf{x}$ toward $\mathbf{r}$ under an admissible factual-counterfactual coupling. This connection can be made precise under a Lipschitz-continuity assumption on the predictive model f . In Theorem 4.1 below (proof in Appendix C), we show that the transportation plan $\mathbf{p}_{\text{OT}}$ used in p -SHAP minimizes an upper bound on the divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$. Specifically, the 1-Wasserstein distance between $f(\mathbf{x})$ and $\mathbf{y}^*$ is bounded by the Lipschitz constant of f multiplied by the square root of the minimized expected cost of transporting $\mathbf{x}$ toward $\mathbf{r}$, namely $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$, where p_{ij} ($j = 1, 2, \dots, m$) encodes how the feature values of $\mathbf{x}_i$ are assigned toward one or more $\mathbf{r}_j$. Operationally, the OT plan $\mathbf{p}_{\text{OT}}$ supplies p -SHAP with a cost-minimizing strategy for adjusting the feature values of $\mathbf{x}$ toward those of $\mathbf{r}$. Relative to any other feasible modification plan ($\mathbf{p} \in \Pi$), $\mathbf{p}_{\text{OT}}$ attains the smallest expected cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$, which correspondingly yields the tightest cost-proportional upper bound on the counterfactual-effect violation $W_1(f(\mathbf{x}), \mathbf{y}^*)$.

Theorem 4.1 (p -SHAP Towards Counterfactual Effect). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{\text{OT}}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi.$$

Equivalently, $\mathbf{p}_{\text{OT}}$ minimizes the expected-feature-modification-cost upper bound on $W_1(f(\mathbf{x}), \mathbf{y}^*)$.

Although Theorem 4.1 concerns transport cost rather than the discrete sparsity objective itself, the transport objective functions as a convex proxy for the $\mathcal{NP}$ -hard L_0 selection problem. By concentrating attribution mass along efficient transport paths, the OT coupling guides the greedy selection in Algorithm 1 toward the elimination of non-essential feature changes. In addition, p -SHAP has a coherent interventional interpretation for attributing behavioral changes to feature modifications, as formalized in Theorem 4.2 below (proof in Appendix D).

Theorem 4.2 (Interventional Effect of p -SHAP). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Furthermore, p -SHAP preserves the standard axiomatic structure enjoyed by B-SHAP and RB-SHAP, which supports its use as a principled feature-attribution mechanism. We omit the proof and refer to (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017) for the relevant Shapley axioms.

5 THE ALGORITHMIC FRAMEWORK COLA

Sketch The algorithmic framework COLA, summarized in Algorithm 1, is designed to solve equation (1) through the coordinated use of four algorithmic components. We describe Algorithm 1 in detail below and provide the accompanying illustration in Figure 2.

Line 1 (Applying a CE algorithm to find a counterfactual $\mathbf{r}$). The CE algorithm A_{CE} receives the model f , the factual data $\mathbf{x}$, the target outcome $\mathbf{y}^*$, and the tolerance ϵ as inputs, and returns a counterfactual $\mathbf{r}$ that remains close to $\mathbf{x}$ while satisfying $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.

Algorithm 1 COUNTERFACTUALS WITH LIMITED ACTIONS (COLA)

Require: Model f , factual $\mathbf{x} \in \mathbb{R}^{n \times d}$, target $\mathbf{y}^* \in \mathbb{R}^m$, ϵ , and C
Ensure: Action plan $\mathbf{c} \in \mathbb{R}^{n \times d}$ and correspondingly a refined counterfactual $\mathbf{z} \in \mathbb{R}^{n \times d}$

- 1: Use $A_{\text{CE}}(f, \mathbf{x}, \mathbf{y}^*, \epsilon)$ to obtain $\mathbf{r} \in \mathbb{R}^{m \times d}$, with $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.
- 2: Use $A_{\text{Prob}}(\mathbf{x}, \mathbf{r})$ to obtain the joint distribution matrix $\mathbf{p} \in \mathbb{R}_+^{n \times m}$.
- 3: Use $A_{\text{Shap}}(\mathbf{x}, \mathbf{r}, \mathbf{p})$ to obtain the shapley values $\phi \in \mathbb{R}^{n \times d}$ for each element of $\mathbf{x}$.
- 4: Normalize the element-wise absolute values of ϕ , i.e., $\varphi_{ik} \leftarrow |\phi_{ik}| / \|\phi\|_1$ ($\varphi \in \mathbb{R}_+^{n \times d}$).
- 5: Use $A_{\text{Value}}(\mathbf{r}, \mathbf{p})$ to obtain matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$.
- 6: For $\mathbf{c} \in \{0, 1\}^{n \times d}$, $c_{ik} \leftarrow 0$ ($i = 1, \dots, n$, $k = 1, \dots, d$).
- 7: Sample C pairs (i, k) according to the probability matrix φ , and let $c_{ik} = 1$ for them.
- 8: Let $\mathbf{z} \leftarrow \mathbf{x}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$).
- 9: **for** $i \leftarrow 1$ to n **do**
- 10: **for** $k \leftarrow 1$ to d **do**
- 11: **if** $c_{ik} = 1$ **then**
- 12: $z_{ik} \leftarrow q_{ik}$
- 13: **end if**
- 14: **end for**
- 15: **end for**
- 16: **return** $\mathbf{c}$ and $\mathbf{z}$

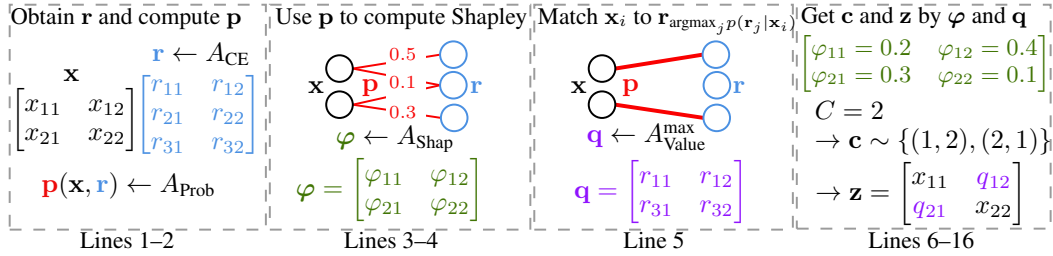


Figure 2: [An illustration of COLA] The figure depicts how COLA constructs $\mathbf{c}$ and $\mathbf{z}$ for equation (1). For expositional simplicity, line 5 is illustrated with $A_{\text{Value}}^{\text{max}}$; in lines 6–16, we set $C = 2$, so sampling from the probability matrix φ produces exactly two modification positions.

Line 2 (*Seeking a joint distribution of $\mathbf{x}$ and $\mathbf{r}$*). The algorithm A_{Prob} takes $\mathbf{x}$ and $\mathbf{r}$ as inputs and produces a matrix representing the joint distribution over all n and m data points in $\mathbf{x}$ and $\mathbf{r}$, respectively. This joint distribution $\mathbf{p}$ encodes an alignment, or matching, between the rows in $\mathbf{x}$ and $\mathbf{r}$, and Line 5 subsequently uses it to compute the candidate values from which $\mathbf{z}$ is assembled. As discussed in Section 3, when row-level alignment is unknown, A_{Prob} can be instantiated via OT so as to obtain the joint distribution with minimal OT distance between $\mathbf{x}$ and $\mathbf{r}$; when such alignment knowledge is available, the joint distribution should instead reflect that known factual–counterfactual correspondence.

Lines 3–4 (*p-SHAP FA*). We apply equation (7) to compute Shapley values for $\mathbf{x}$. The joint distribution $\mathbf{p}$ may be used to align each row of $\mathbf{x}$ with the corresponding counterfactual rows of $\mathbf{r}$, so that those selected counterfactual rows function as the most representative contrastive references for the row in $\mathbf{x}$. Numerically, this alignment has a substantial effect on contrastive FA. The resulting Shapley-value matrix for $\mathbf{x}$ is then transformed elementwise by absolute value and normalized so that its entries sum to one; the normalized matrix φ constitutes the attribution policy used for selecting modifications.

Line 5 (*Computing feature values*). The value-assignment algorithm A_{Value} receives the counterfactual set $\mathbf{r}$ and the joint distribution $\mathbf{p}$, and, for each row i of $\mathbf{x}$, the algorithm A_{Value} selects one or more rows of $\mathbf{r}$ as references for modifying $\mathbf{x}$ toward $\mathbf{r}$. It returns a matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$ in which each entry q_{ik} is a candidate counterfactual replacement for x_{ik} ($\forall i, k$). We consider $A_{\text{Value}}^{\text{max}}$ for single-row selection and $A_{\text{Value}}^{\text{avg}}$ for multi-row aggregation.

For any row $\mathbf{x}_i$, $A_{\text{Value}}^{\max}$ selects the row of $\mathbf{r}$ assigned the largest probability mass.

$$\mathbf{q} = A_{\text{Value}}^{\max}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = r_{\tau(i),k} \text{ and } \tau(i) = \arg \max_{j=1,2,\dots,m} p_{ij}. \quad (9)$$

The algorithm A^{avg} instead computes q_{ik} as a convex combination, namely a probability-weighted average of $r_{1k}, r_{2k}, \dots, r_{mk}$.

$$\mathbf{q} = A_{\text{Value}}^{\text{avg}}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}. \quad (10)$$

Lines 6–16. Because the non-negative matrix φ is normalized to sum to one, it can be interpreted as a stochastic policy for choosing entries of $\mathbf{x}$ to replace, i.e., entries for which $c_{ik} = 1$, as implemented in line 7. For every selected pair (i, k) , the factual value x_{ik} is replaced by q_{ik} ; the resulting modified matrix is returned as $\mathbf{z}$, together with $\mathbf{c}$, as the optimized solution to equation (1).

By Theorem 4.1, COLA is structured to reduce the dissimilarity between $f(\mathbf{z})$ and $\mathbf{y}^*$ by using feature-attribution scores to identify the entries whose adjustment is most relevant for the desired counterfactual effect. The theorem below (proof in Appendix E) shows that the refined $\mathbf{z}$, produced as the final $\mathbf{z}$ by COLA, satisfies the proximity requirement in equation (1b) under the Frobenius-norm formulation. Empirical evidence supporting the broader practical relevance of this conclusion appears in Table 3.

Theorem 5.1 (Counterfactual Proximity). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{Value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is no farther from the factual input $\mathbf{x}$ than the aligned counterfactual proposal $\mathbf{q}$, and therefore that refinement does not introduce additional divergence.

Corollary 5.2. *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

thereby recovering the intuition that the refined counterfactual is at least as close to the factual observations as the original counterfactual set after reordering by σ .

Complexity of COLA Let $\mathcal{O}(M_{\text{CE}})$ denote the computational complexity of A_{CE} . For A_{Shap} , suppose that weighted linear regression is used to estimate Shapley values, and let M_{Shap} denote the number of sampled subsets. The resulting complexity of COLA, expressed in terms of n, m, d , and the entropic-OT regularization parameter ε , is $\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N$ where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ when $A_{\text{Value}}^{\max}$ is used and $N = \mathcal{O}(nmd)$ when $A_{\text{Value}}^{\text{avg}}$ is used. Appendix F provides the corresponding derivation.

6 NUMERICAL RESULTS

This section evaluates the effectiveness of COLA for the optimization problem in equation (1), setting $\mathbf{y}^* = f(\mathbf{r})$ where $\mathbf{r}$ is produced by a CE method A_{CE} . We use four divergence functions in order to assess distinct notions of discrepancy: OT measures distribution-level distance, MMD measures divergence between distributions through mean embeddings in a high-dimensional feature space, and absolute mean difference (MeanD) and absolute median difference (MedianD) measure discrepancies between means and medians, respectively. The numerical evaluation is organized around three claims: I) COLA’s effectiveness for modification minimality. II) p -SHAP’s superior performance than other Shapley methods towards modification minimality. III) COLA’s near-optimal performance.

Table 1: [Experiment Scenarios Setup] Four datasets support the benchmarking of five CE algorithms across 12 predictive models. Each “scenario” corresponds to one combination of dataset, A_{CE} algorithm, and model f .

Dataset	HELOC (FICO, 2018), German Credit (Hofmann, 1994), Hotel Bookings (Antonio et al., 2019), COMPAS (Jeff Larson et al., 2016)
A_{CE}	DiCE (Mothilal et al., 2020), AReS (Rawal & Lakkaraju, 2020), GlobeCE (Ley et al., 2023), KNN (Albini et al., 2022; Contardo et al.; Forel et al., 2023), Discount (You et al., 2025)
Model f	Bagging, LightGBM, Support Vector Machine (SVM), Gaussian Process (GP), Radial Basis Function Network (RBF), XGBoost, Deep Neural Network (DNN), Random Forest (RndForest), AdaBoost, Gradient Boosting (GradBoost), Logistic Regression (LR), Quadratic Discriminant Analysis (QDA)

Table 2: [Experiment Methods Setup] The table defines the 6 comparison methods, with colors matched to Figure 3. Each method is evaluated within the experiment scenarios specified in Table 1.

Method	The probability $\mathbf{p}$ used by A_{value} and A_{Shap}	The Shapley algorithm A_{Shap}
RB- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
RB- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8) (but not used in } A_{\text{Shap}})$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
CF- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{Rnd}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched randomly to an } \mathbf{r}_j$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8)}$	p -SHAP with $\mathbf{p}$
CF- p_{Ect}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched to known counterpart } \mathbf{r}_j$	(CF or B)-SHAP, $\mathcal{D}(\mathbf{x}_i) \rightarrow \mathbf{r}_j$

Experiment Setup The experiments use 4 binary-classification datasets, 5 CE algorithms designed for heterogeneous objectives, and 12 classifiers, as summarized in Table 1. We define an “experiment scenario” as a particular combination of dataset, A_{CE} algorithm, and predictive model.

We briefly characterize the A_{CE} methods in Table 1. DiCE and KNN are instance-based CE methods that return counterfactuals for individual factual observations, whereas AReS and GlobeCE are group-based methods that identify counterfactual collections for the factual data considered jointly. Discount adopts a distributional perspective, treating the factual instances as an empirical distribution and seeking a nearby counterfactual distribution.

Table 2 defines six methods, among which CF- p_{OT} is the proposed p -SHAP instantiation and the remaining methods serve as baselines. Each method is evaluated across the experiment scenarios in Table 1. Method specification is determined by the triplet of algorithms A_{Prob} , A_{value} , and A_{Shap} ; for instance, RB- p_{Uni} uses a uniform distribution within A_{Prob} to compute $\mathbf{p}$, passes the resulting $\mathbf{p}$ through either equation (9) or equation (10) in A_{value} to compute $\mathbf{q}$, and uses RB-SHAP as A_{Shap} .

The comparison methods are designed to isolate the contribution of each modeling choice. First, RB- p_{Uni} and CF- p_{Uni} differ only in whether CE information is used, with the latter incorporating counterfactual information and the former omitting it. Second, CF- p_{Uni} , CF- p_{Rnd} , and CF- p_{OT} all use CE information in FA, but they impose different joint distributions; the results below show that the OT-derived distribution is substantially more effective. Third, comparing CF- p_{OT} , which uses the OT alignment inside A_{Shap} , with RB- p_{OT} , which does not, isolates whether the benefit of OT comes specifically from factual-counterfactual alignment rather than from unrelated design choices. Finally, CF- p_{Ect} represents the special case in which each counterfactual has a known source, so exact factual-counterfactual alignment is available and CF-SHAP coincides with B-SHAP.

Result I: COLA achieves significant action reduction with a minor loss in counterfactual effect In Table 3, we set the goal for $\mathbf{z}$ such that the refined $\mathbf{z}$ reaches either 80% or 100% of the counterfactual effect achieved by $f(\mathbf{r})^2$. The results show that COLA attains these targets while requiring substantially fewer actions in $\mathbf{z}$, i.e., fewer feature modifications in $\mathbf{z}$, than the original CE

²Note that by definition, we have $D(f(\mathbf{r}), \mathbf{y}^*) = 0$, which represents a 100% counterfactual effect since $\mathbf{y}^* = f(\mathbf{r})$. To define a counterfactual effect 80%, consider the proportion of divergence reduced by the refined CE $\mathbf{z}$. That is, $\text{Counterfactual Effect} = 1 - D(f(\mathbf{z}), \mathbf{y}^*)/D(f(\mathbf{x}), \mathbf{y}^*) = 80\%$, with D being OT.

Table 3: [COLA for Modification Minimality] The table reports, for each method, the number of modified features in $\mathbf{z}$ required for $f(\mathbf{z})$ to attain 80% of the counterfactual effect of $f(\mathbf{r})$. Each entry averages results over 4 randomly selected scenarios from Table 1, with 100 runs per scenario; “-” indicates that the target counterfactual effect is not achieved.

Dataset	Method	80% Counterfactual Effect		100% Counterfactual Effect	
		# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$	# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$
German Credit $ \mathcal{F} = 9$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	5.29(± 0.09)	75.9%	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	—	—	—	—
	CF- p_{OT}	1.70(± 0.02)	24.3%	3.13(± 0.03)	44.9%
Hotel Bookings $ \mathcal{F} = 29$	RB- p_{Uni}	7.10(± 0.08)	24.5%	—	—
	RB- p_{OT}	8.55(± 0.08)	50.2%	—	—
	CF- p_{Uni}	7.01(± 0.07)	41.1%	—	—
	CF- p_{Rnd}	10.63(± 0.08)	62.4%	—	—
	CF- p_{OT}	2.50(± 0.03)	14.6%	4.44(± 0.02)	26.0%
COMPAS $ \mathcal{F} = 15$	RB- p_{Uni}	5.02(± 0.05)	82.7%	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	2.80(± 0.04)	34.4%	—	—
	CF- p_{Rnd}	2.58(± 0.04)	32.1%	—	—
	CF- p_{OT}	1.25(± 0.03)	14.8%	2.45(± 0.03)	30.0%
HELOC $ \mathcal{F} = 23$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	2.73(± 0.04)	15.7%	—	—
	CF- p_{OT}	2.35(± 0.03)	13.4%	7.745(± 0.05)	44.7%

$\mathbf{r}$. To obtain an 80% counterfactual effect, COLA requires only 13%–25% of the feature changes in the original counterfactual, as measured by $\|\mathbf{z} - \mathbf{x}\|/\|\mathbf{r} - \mathbf{x}\|$. Moreover, only COLA with p -SHAP, namely CF- p_{OT} , reaches the 100% counterfactual-effect target, doing so with only 26%–45% of the feature changes in the original $\mathbf{r}$. Thus, p -SHAP yields the strongest action minimality among the evaluated methods, as analyzed in greater detail below.

Result II: p -SHAP outperforms the other Shapley methods in achieving counterfactual effect

Figure 3 evaluates the Shapley methods in equation (4)–equation (7), with the x-axis denoting the allowed number of feature changes C and the y-axis reporting $D(f(\mathbf{z}), \mathbf{y}^*)$ from equation (1). First, RB- p_{Uni} and RB- p_{OT} perform substantially worse than the counterfactual-informed alternatives, indicating that CE information is crucial when attribution is used to achieve modification minimality in CE. Second, the superiority of CF- p_{OT} over RB- p_{OT} shows that OT improves p -SHAP specifically through factual–counterfactual alignment rather than through unrelated factors, since these two methods differ only in A_{Shap} (see Table 2). Third, p -SHAP substantially outperforms CF- p_{Uni} and CF- p_{Rnd} , demonstrating that the joint distribution induced by OT is essential: simply incorporating counterfactual information into FA, as in CF- p_{Uni} and CF- p_{Rnd} , is insufficient unless an appropriate factual–counterfactual alignment is also imposed, although that alignment need not coincide with the exact generation mechanism, as Figure 4 later illustrates.

Across many of the experiment scenarios defined in Table 1, COLA with p -SHAP exhibits strong performance, as shown in Figure 3 and Table 3. Additional results of the same form are reported in Appendix H. Overall, p -SHAP outperforms the alternative Shapley methods for inducing sparsity in CE.

Result III: COLA is competitive with the mixed integer linear programming (MILP) optimum on tractable benchmarks

This result demonstrates that p -SHAP can attenuate dependence on the CE generation process by replacing CE algorithm-dependent knowledge³ of $\mathcal{D}$ with the OT joint distribution between factual and counterfactual data, as shown in Figure 4. We benchmark CF- p_{Ect}

³This knowledge, i.e. exact alignment between factual and counterfactual, is available only by DiCE and KNN among all CE methods considered.

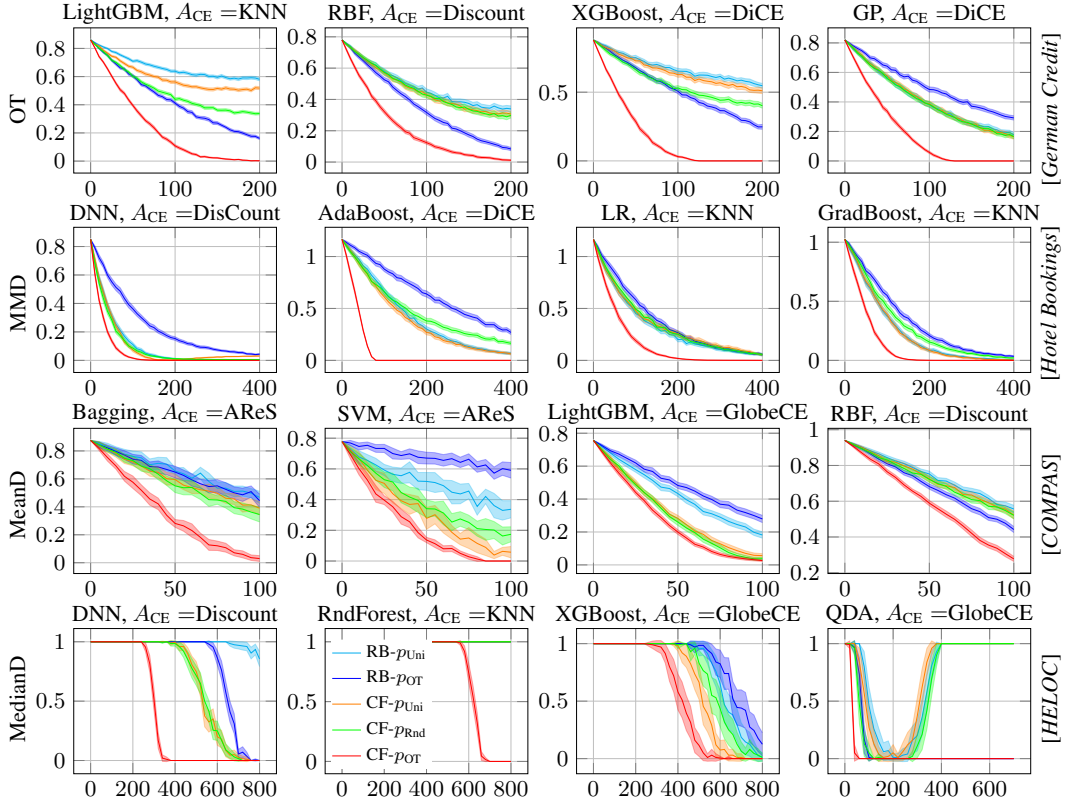


Figure 3: $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the allowed action budget C , averaged over 100 runs. Shaded regions indicate 99.9% confidence intervals. $A_{\text{Value}}^{\text{avg}}$ is used for *HELOC* and *COMPAS*, whereas $A_{\text{Value}}^{\text{max}}$ is used for *German Credit* and *Hotel Bookings*. The legend inside “RnDForest, $A_{\text{CE}} = \text{KNN}$ ” applies to all panels.

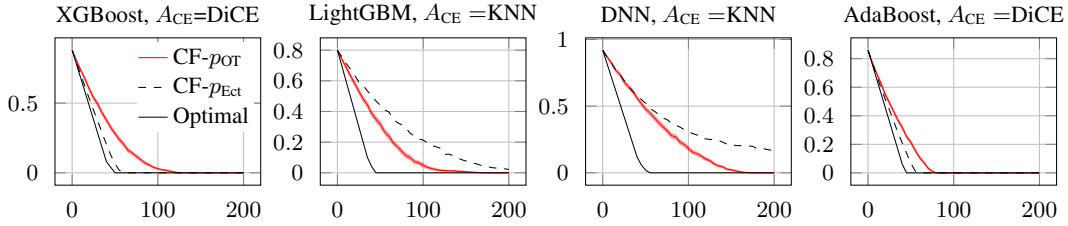


Figure 4: [*German Credit*] $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the allowed action budget C , with D instantiated as *MeanD*.

within COLA and focus on solving equation (1) with *MeanD* as the divergence function D . Because $\text{CF-}p_{\text{Ect}}$ relies on known factual–counterfactual alignment, this comparison evaluates how effectively COLA uses such alignment information. The theoretical optimum for this case can be obtained by solving an MILP; Appendix G derives the corresponding formulation. Since solving the MILP is computationally demanding, this benchmark is restricted to *German Credit*.

The results show that $\text{CF-}p_{\text{Ect}}$ attains near-optimal performance with DiCE. Because the factual–counterfactual pairs produced by DiCE and KNN are independent across observations, these findings indicate that COLA is effective for instance-based CE, even though the formulation in equation (1) generalizes to group and distributional CE. Notably, $\text{CF-}p_{\text{OT}}$ sometimes outperforms $\text{CF-}p_{\text{Ect}}$, because it utilizes a more theoretically grounded approach to identify the key features that require modification, whereas $\text{CF-}p_{\text{Ect}}$ relies on CE algorithm-dependent knowledge, which lacks solid justification on its

effectiveness for FA. Nevertheless, a gap remains between $CF\text{-}p_{\text{Ect}}$, $CF\text{-}p_{\text{OT}}$, and the KNN optimum, so identifying the best alignment remains an open question.

7 CONCLUSIONS

This paper introduced COLA, a post-hoc framework for refining CE through joint-distribution-informed Shapley values, with the objective of preserving the counterfactual effect while substantially reducing the number of required actions. By coupling factual and counterfactual observations through OT and using the resulting alignment to guide attribution and value replacement, COLA provides a model- and generator-agnostic mechanism for converting existing counterfactuals into sparser, more actionable explanations without altering the underlying experimental claims or counterfactual targets.

ETHICS STATEMENT

All authors have read and agree to abide by the ICLR Code of Ethics.⁴ The paper proposes a methodological and empirical approach, namely *COLA* with *p*-SHAP, for refining CE; it involves neither human-subject participation nor real-world interventions, and it does not collect, annotate, or release new personal data.

All experiments rely on standard, de-identified, publicly available tabular datasets, such as Adult/Census Income, German Credit, and HELOC. We follow the applicable dataset licenses and terms, use the data solely for benchmarking in accordance with common research practice, make no attempt at re-identification or record linkage, and release no additional personal information. The accompanying code will include mechanisms for excluding protected attributes and enforcing feasibility or immutability constraints on features.

Although CE can expose biases in the underlying predictive models, they do not by themselves guarantee fairness. Our method reduces the number of recommended feature edits, but it does *not* authorize changes to sensitive or immutable attributes (e.g., sex, race, age), nor does it establish causal validity. We discourage uses of CE that would enable attempts to “game” high-stakes systems, such as lending, hiring, or healthcare, or that would pressure individuals toward unrealistic or unethical behavioral changes. Any deployment in consequential domains should incorporate domain-expert-defined feasibility constraints, fairness audits, including assessments of disparate impact or benefit across groups, human oversight, and compliance with relevant data-protection and AI regulations. We explicitly caution that the attribution is distributional rather than causal and must not be interpreted as a causal effect.

The principal risks are (i) misinterpretation of attributions as causal claims, (ii) recommendation of infeasible or harmful edits, and (iii) facilitation of strategic manipulation. We mitigate these risks by explicitly delimiting the non-causal scope and limitations of the method, supporting cost and feasibility masks as well as immutable-feature constraints in the implementation, reporting subgroup performance where relevant, and providing responsible-use guidance in the documentation.

To facilitate replication, we provide an anonymized implementation together with detailed settings and random seeds; we report model classes, hyperparameters, and compute budgets, and we reduce the risk of result cherry-picking through fixed evaluation protocols. To our knowledge, this submission involves no conflicts of interest or sensitive sponsorships, and any funding disclosures will be supplied after double-blind review.

REPRODUCIBILITY STATEMENT

We treat reproducibility as a central requirement and therefore identify the components needed to reconstruct the study. The COLA framework and *p*-SHAP attribution are specified through notation and pseudocode in the main paper, including Algorithm 1 and the Method and Preliminaries sections, while theoretical assumptions and complete proofs are deferred to the appendix, where theorems and lemmas are accompanied by line-by-line arguments.

⁴<https://iclr.cc/public/CodeOfEthics>

Appendix I documents the implementation details, including hyperparameters, optimization settings, environment specifications, and random seeds, and also specifies the datasets and evaluation protocol, including data splits and preprocessing. The experimental sections and ablations enumerate the baselines, including predictive models and CE generators, as well as the coupling choices, with corresponding configurations mirrored in the supplemental materials. The experiment code and software will be released upon acceptance; Appendix I also reports hardware and runtime to contextualize compute requirements, and the appendix provides ablation and sensitivity studies together with the small-scale MILP benchmark setup to support independent verification.

REFERENCES

- Google OR-Tools. URL <https://developers.google.com/optimization>.
- Gurobi Optimization, LLC. URL <https://www.gurobi.com>.
- Machine Learning in Python. URL <https://scikit-learn.org/stable/>.
- Emanuele Albini, Jason Long, Danial Dervovic, and Daniele Magazzeni. Counterfactual shapley additive explanations. In *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, pp. 1054–1070, 2022.
- Nuno Antonio, Ana de Almeida, and Luis Nunes. Hotel booking demand datasets. *Data in brief*, 22: 41–49, 2019.
- Alejandro Barredo Arrieta, Natalia Díaz-Rodríguez, Javier Del Ser, Adrien Bennetot, Siham Tabik, Alberto Barbado, Salvador García, Sergio Gil-López, Daniel Molina, Richard Benjamins, et al. Explainable artificial intelligence (xai): Concepts, taxonomies, opportunities and challenges toward responsible ai. *Information fusion*, 58:82–115, 2020.
- Emilio Carrizosa, Jasone Ramírez-Ayerbe, and Dolores Romero Morales. Mathematical optimization modelling for group counterfactual explanations. *European Journal of Operational Research*, 2024.
- Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, pp. 785–794, 2016.
- Claudio Contardo, Ricardo Fukasawa, Louis-Martin Rousseau, and Thibaut Vidal. Optimal counterfactual explanations for k-nearest neighbors using mathematical optimization and constraint programming. URL <https://optimization-online.org/>.
- Arun Das and Paul Rad. Opportunities and challenges in explainable artificial intelligence (xai): A survey. *arXiv preprint arXiv:2006.11371*, 2020.
- FICO. Fico explainable machine learning challenge, 2018. URL <https://community.fico.com/s/explainable-machine-learning-challenge>.
- Alexandre Forel, Axel Parmentier, and Thibaut Vidal. Explainable data-driven optimization: From context to decision and back again. In *International Conference on Machine Learning*, pp. 10170–10187. PMLR, 2023.
- Riccardo Guidotti. Counterfactual explanations and how to find them: literature review and benchmarking. *Data Mining and Knowledge Discovery*, pp. 1–55, 2022.
- Hans Hofmann. Statlog (German Credit Data). UCI Machine Learning Repository, 1994. DOI: <https://doi.org/10.24432/C5NC77>.
- Lauren Kirchner Jeff Larson, Surya Mattu, and J Angwin. How we analyzed the compas recidivism algorithm, 2016. URL <https://www.propublica.org/article/how-we-analyzed-the-compas-recidivism-algorithm>.
- Kentaro Kanamori, Takuya Takagi, Ken Kobayashi, and Hiroki Arimura. Dace: Distribution-aware counterfactual explanation by mixed-integer linear optimization. In *IJCAI*, pp. 2855–2862, 2020.

648 Kentaro Kanamori, Takuya Takagi, Ken Kobayashi, and Yuichi Ike. Counterfactual explanation trees:
649 Transparent and consistent actionable recourse with decision trees. In *International Conference on*
650 *Artificial Intelligence and Statistics*, pp. 1846–1870. PMLR, 2022.

651 Amir-Hossein Karimi, Bernhard Schölkopf, and Isabel Valera. Algorithmic recourse: from coun-
652 terfactual explanations to interventions. In *Proceedings of the 2021 ACM conference on fairness,*
653 *accountability, and transparency*, pp. 353–362, 2021.

654 Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan
655 Liu. Lightgbm: A highly efficient gradient boosting decision tree. *Advances in neural information*
656 *processing systems*, 30, 2017.

657 Ramaravind Kommiya Mothilal, Divyat Mahajan, Chenhao Tan, and Amit Sharma. Towards uni-
658 fying feature attribution and counterfactual explanations: Different means to the same end. In
659 *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society*, pp. 652–663, 2021.

660 Yongchan Kwon and James Y Zou. Weightedshap: analyzing and improving shapley based feature
661 attributions. *Advances in Neural Information Processing Systems*, 35:34363–34376, 2022.

662 Dan Ley, Saumitra Mishra, and Daniele Magazzeni. Global counterfactual explanations: Investiga-
663 tions, implementations and improvements. *arXiv preprint arXiv:2204.06917*, 2022.

664 Dan Ley, Saumitra Mishra, and Daniele Magazzeni. Globe-ce: a translation based approach for global
665 counterfactual explanations. In *International Conference on Machine Learning*, pp. 19315–19342.
666 PMLR, 2023.

667 Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions. *Advances in*
668 *neural information processing systems*, 30, 2017.

669 Luke Merrick and Ankur Taly. The explanation game: Explaining machine learning models using
670 shapley values. In *Machine Learning and Knowledge Extraction: 4th IFIP TC 5, TC 12, WG*
671 *8.4, WG 8.9, WG 12.9 International Cross-Domain Conference, CD-MAKE 2020, Dublin, Ireland,*
672 *August 25–28, 2020, Proceedings 4*, pp. 17–38. Springer, 2020.

673 Ramaravind K Mothilal, Amit Sharma, and Chenhao Tan. Explaining machine learning classifiers
674 through diverse counterfactual explanations. In *Proceedings of the 2020 conference on fairness,*
675 *accountability, and transparency*, pp. 607–617, 2020.

676 Yasunobu Nohara, Koutarou Matsumoto, Hidehisa Soejima, and Naoki Nakashima. Explanation
677 of machine learning models using shapley additive explanation and application for real data in
678 hospital. *Computer Methods and Programs in Biomedicine*, 214:106584, 2022.

679 Axel Parmentier and Thibaut Vidal. Optimal counterfactual explanations in tree ensembles. In
680 *International conference on machine learning*, pp. 8422–8431. PMLR, 2021.

681 Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor
682 Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. Pytorch: An imperative style,
683 high-performance deep learning library. *Advances in neural information processing systems*, 32,
684 2019.

685 Kaivalya Rawal and Himabindu Lakkaraju. Beyond individualized recourse: Interpretable and
686 interactive summaries of actionable recourses. *Advances in Neural Information Processing Systems*,
687 33:12187–12198, 2020.

688 Amit Sharma, Hua Li, and Jian Jiao. The counterfactual-shapley value: Attributing change in system
689 metrics. In *NeurIPS 2022 Workshop on Causality for Real-world Impact*.

690 Yi Sun and Mukund Sundararajan. Axiomatic attribution for multilinear functions. In *Proceedings of*
691 *the 12th ACM conference on Electronic commerce*, pp. 177–178, 2011.

692 Mukund Sundararajan and Amir Najmi. The many shapley values for model explanation. In
693 *International conference on machine learning*, pp. 9269–9278. PMLR, 2020.

702 Lucile Ter-Minassian, Sahra Ghalebikesabi, Karla Diaz-Ordaz, and Chris Holmes. Challenges and
703 opportunities of shapley values in a clinical context. *arXiv preprint arXiv:2306.14698*, 2023.
704

705 Berk Ustun, Alexander Spangher, and Yang Liu. Actionable recourse in linear classification. In
706 *Proceedings of the conference on fairness, accountability, and transparency*, pp. 10–19, 2019.

707 Sahil Verma, John Dickerson, and Keegan Hines. Counterfactual explanations for machine learning:
708 A review. *arXiv preprint arXiv:2010.10596*, 2, 2020.
709

710 Sandra Wachter, Brent Mittelstadt, and Chris Russell. Counterfactual explanations without opening
711 the black box: Automated decisions and the gdpr. *Harv. JL & Tech.*, 31:841, 2017.

712 Lei You, Lele Cao, Mattias Nilsson, Bo Zhao, and Lei Lei. Distributional counterfactual explanations
713 with optimal transport. In *The 28th International Conference on Artificial Intelligence and Statistics*,
714 2025.
715
716
717
718
719
720
721
722
723
724
725
726
727
728
729
730
731
732
733
734
735
736
737
738
739
740
741
742
743
744
745
746
747
748
749
750
751
752
753
754
755

APPENDICES

A	Comparison to Existing Approaches	15
B	$\mathcal{NP}$-hardness of the Problem in equation (1)	15
C	Proof of Theorem 4.1: p-SHAP Towards Counterfactual Effect	16
D	Proof of Theorem 4.2: Interventional Effect	18
E	Proof of Theorem 5.1: Counterfactual Proximity	19
F	Analysis of Computational Complexity of COLA	20
G	An MILP Formulation of equation (1) With MeanD	20
H	Extended Numerical Results	21
I	Experiments Reproducibility	22

A COMPARISON TO EXISTING APPROACHES

(Albini et al., 2022) proposed CF-SHAP, which uses counterfactual data points as the Shapley background distribution. The method, however, assumes that the counterfactual distribution is defined conditionally for each individual factual instance, thereby implying the availability of a probabilistic alignment between factual and counterfactual observations. Related assumptions also appear in (Merrick & Taly, 2020; Kommiya Mothilal et al., 2021). In many scenarios where global explanations are expected, this assumption fails. We note that the setup in (Albini et al., 2022) for counterfactual data distribution is a special case of ours. The authors in (Kwon & Zou, 2022) proposed WeightedSHAP, adding weights to features rather than treating them as contributing equally. Our proposed method weights the contributions for data points and can be straightforwardly extended to consider weighting both rows and columns. Literature (Kommiya Mothilal et al., 2021) establishes a framework for utilizing both FA and CE for explainability. Yet, the CE-based FA have the same assumption as (Albini et al., 2022), making it difficult to generalize to group (Ley et al., 2023; 2022) or distributional CE (You et al., 2025) cases. Another relevant work is (Sharma et al.), a method that models counterfactuals using a system’s causal structure and time-series predictors to fairly decompose a single observed change in a system-level metric into contributions from individual inputs. More importantly, the aforementioned literature does not address the minimal actions CE problem, which is the focus of our paper.

The problems formulated in (Kanamori et al., 2022; Karimi et al., 2021) are quite close to the one investigated in this paper. In (Karimi et al., 2021) the authors minimize the cost of performing actions with assumption of known structural causal model (SCM), which is rarely known in practice. The authors in (Kanamori et al., 2022) pointed out that it remains open whether existing CE methods can be used for solving that problem. An-MILP-solvers based approach is proposed for linear classifiers, tree ensembles, and deep ReLU networks, built upon the works (Ustun et al., 2019; Kanamori et al., 2020; Parmentier & Vidal, 2021). However, solving MILP is costly, which makes it difficult to scale.

B $\mathcal{NP}$ -HARDNESS OF THE PROBLEM IN EQUATION (1)

Theorem B.1 establishes that, unless $\mathcal{P} = \mathcal{NP}$, one should not expect a generally scalable exact algorithm for this problem; importantly, the hardness is not merely a consequence of nonlinear predictive models, but also of the combinatorial structure induced by sparse action selection.

Theorem B.1. *Problem equation (1) is, for non-trivial divergences, generally $\mathcal{NP}$ -hard; in particular, hardness persists even when $d = 1$ and the predictive model is linear.*

Proof. Consider the *Sparse Regression (SR) with a Cardinality Constraint* problem, defined as follows. Given a matrix $\mathbf{W} \in \mathbb{R}^{m \times n}$, a target vector $\mathbf{y}^* \in \mathbb{R}^m$, and a sparsity level $K \in \mathbb{N}$, the goal is to find a vector $\mathbf{z} \in \mathbb{R}^n$ that minimizes the residual error while having at most K non-zero

elements:

$$\begin{aligned} \min_{\mathbf{z}} \quad & \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \\ \text{s.t.} \quad & \|\mathbf{z}\|_0 \leq K, \end{aligned}$$

where $\|\mathbf{z}\|_0$ denotes the number of non-zero elements in $\mathbf{z}$. This problem is known to be $\mathcal{NP}$ -hard due to the combinatorial nature of selecting the subset of variables to include in the model.

We map this SR instance to Problem equation (1) through the following construction. Set the number of features to $d = 1$ and the number of instances to n , matching the dimension of the SR problem; take the factual data to be $\mathbf{x} = \mathbf{0} \in \mathbb{R}^n$, and use the given SR target as $\mathbf{y}^* \in \mathbb{R}^m$. Finally, define the predictive model as the linear map $f(\mathbf{z}) = \mathbf{W}\mathbf{z}$.

For model outputs, define $D(f(\mathbf{z}), \mathbf{y}^*) = \|f(\mathbf{z}) - \mathbf{y}^*\|_2^2$, the squared Euclidean distance. For instances, define $D(\mathbf{z}, \mathbf{x}) = \|\mathbf{z} - \mathbf{x}\|_2^2 = \|\mathbf{z}\|_2^2$, using $\mathbf{x} = \mathbf{0}$. The maximum number of allowed feature changes C is set equal to k , the sparsity level in the SR instance, and the big- M constant may be any sufficiently large positive number, for example $M \geq \max_i |z_i|$.

Given that $\mathbf{x} = \mathbf{0}$, constraints equation (1d) and equation (1e) are simplified to:

$$\begin{aligned} z_i &\leq 0 \cdot (1 - c_i) + M c_i = M c_i, \\ z_i &\geq 0 \cdot (1 - c_i) - M c_i = -M c_i, \quad \forall k = 1, \dots, n. \end{aligned}$$

Thus, when $c_i = 0$, the two inequalities force $z_i \leq 0$ and $z_i \geq 0$, and hence $z_i = 0$; when $c_i = 1$, the variable is allowed to range over $z_i \in [-M, M]$. Constraint equation (1c) correspondingly reduces to $\sum_{k=1}^n c_i \leq k$.

Our problem equation (1) thus becomes:

$$\min_{\mathbf{c}, \mathbf{z}} \quad \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \tag{11a}$$

$$\text{s.t.} \quad \|\mathbf{z}\|_2^2 \leq \epsilon, \tag{11b}$$

$$\sum_{k=1}^d c_i \leq k, \tag{11c}$$

$$z_i = 0, \quad \text{if } c_i = 0, \tag{11d}$$

$$z_i \in [-M, M], \quad \text{if } c_i = 1, \tag{11e}$$

$$c_i \in \{0, 1\}, \quad \forall k = 1, \dots, n. \tag{11f}$$

In this formulation, c_i indicates whether z_i is permitted to change: $c_i = 1$ allows modification, whereas $c_i = 0$ enforces $z_i = 0$, thereby reproducing the sparsity mechanism of the SR problem. The constraint $\sum_{k=1}^n c_i \leq k$ ensures that no more than k entries can be active, exactly matching the prescribed sparsity level, and the objective coincides with the SR objective.

Because this construction directly embeds the cardinality-constrained SR problem, which is known to be $\mathcal{NP}$ -hard, solving Problem equation (1) is at least as hard as solving that SR instance. Consequently, Problem equation (1) remains $\mathcal{NP}$ -hard even with $d = 1$, a linear model f , and standard Euclidean divergences D . $\square$

C PROOF OF THEOREM 4.1: p -SHAP TOWARDS COUNTERFACTUAL EFFECT

Let $\mathbf{x} = \{\mathbf{x}_i\}_{i=1}^n \in \mathbb{R}^{n \times d}$ be the set of factual data points with associated probability weights $\mu_i \geq 0$ such that $\sum_{i=1}^n \mu_i = 1$, and let $\mathbf{r} = \{\mathbf{r}_j\}_{j=1}^m \in \mathbb{R}^{m \times d}$ be the set of counterfactual data points with associated probability weights $\nu_j \geq 0$ such that $\sum_{j=1}^m \nu_j = 1$. Let $\mathbf{p}_{\text{OT}} \in \mathbb{R}^{n \times m}$ be the OT plan between $\mathbf{x}$ and $\mathbf{r}$ that minimizes the expected transportation cost:

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2,$$

where $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ is the set of joint distributions satisfying the marginal constraints $\sum_{j=1}^m p_{ij} = \mu_i$ for all i and $\sum_{i=1}^n p_{ij} = \nu_j$ for all j . The theorem below shows that the resulting feature attributions are aligned with expected feature-modification costs, thereby inducing action plans that are cost-efficient for attaining counterfactual outcomes.

Theorem C.1 (Theorem 4.1 in the main text). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{OT}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \quad \forall \mathbf{p} \in \Pi.$$

Equivalently, $\mathbf{p}_{OT}$ minimizes the expected-feature-modification-cost upper bound on $W_1(f(\mathbf{x}), \mathbf{y}^*)$.

Proof. Since the model f is Lipschitz continuous with constant L , for any $\mathbf{x}_i \in \mathbb{R}^d$ and $\mathbf{r}_j \in \mathbb{R}^d$, it holds that:

$$|f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Multiplying both sides of the inequality by $p_{ij}^{OT} \geq 0$, we obtain:

$$p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Summing both sides over all $i = 1, \dots, n$ and $j = 1, \dots, m$, we have:

$$\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Let us denote $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$ and $E_d = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2$. The inequality then becomes:

$$E_f \leq L E_d.$$

To further bound E_d , we apply the Cauchy-Schwarz inequality. Observe that the weights p_{ij}^{OT} are non-negative and satisfy $\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} = 1$ because $\mathbf{p}_{OT}$ is a probability distribution over the joint space of $\mathbf{x}$ and $\mathbf{r}$. The Cauchy-Schwarz inequality states that for any real-valued functions a_{ij} and b_{ij} ,

$$\left(\sum_{i,j} a_{ij} b_{ij} \right)^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right).$$

Setting $a_{ij} = \sqrt{p_{ij}^{OT}}$ and $b_{ij} = \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2$, we have:

$$E_d = \sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} \sqrt{p_{ij}^{OT}} \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} a_{ij} b_{ij}.$$

Applying the Cauchy-Schwarz inequality:

$$E_d^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right) = \left(\sum_{i,j} p_{ij}^{OT} \right) \left(\sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 \right).$$

Since $\sum_{i,j} p_{ij}^{OT} = 1$, this simplifies to:

$$E_d^2 \leq \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2.$$

Taking the square root of both sides, we obtain:

$$E_d \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Substituting back into the inequality for E_f , we have:

$$E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Note that the 1-Wasserstein divergence is no more than E_f^5 , then,

$$W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*| \leq E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Therefore, the 1-Wasserstein divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$ is controlled by the Lipschitz constant L multiplied by the square root of the expected transportation cost induced by $\mathbf{p}_{\text{OT}}$. Since $\mathbf{p}_{\text{OT}}$ minimizes the expected transportation cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$ over all feasible transport plans in $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$, we have

$$\sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

Hence the conclusion follows. $\square$

D PROOF OF THEOREM 4.2: INTERVENTIONAL EFFECT

Theorem D.1 (Theorem 4.2 in the main text). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Proof. Let $p(\mathbf{x}, \mathbf{r})$ be the joint probability of $\mathbf{x}$ and $\mathbf{r}$ obtained from A_{Prob} . Under the intervention $do(\mathbf{r}_S = \mathbf{x}_{i,S})$, the features in S are set to $\mathbf{x}_{i,S}$, and the features in $\mathcal{F} \setminus S$ remain distributed according to their marginal distribution $p(\mathbf{r}_{\mathcal{F} \setminus S})$. Therefore, the expected value of $f(\mathbf{r})$ under the intervention is:

$$\mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S}.$$

Remark that by the definition of $v^{(i)}(S)$,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r} | \mathbf{x}_i)}[f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}[f(\mathbf{r})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S} - \mathbb{E}[f(\mathbf{r})],$$

such that

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Hence the conclusion. $\square$

The value function $v^{(i)}(S)$ captures the expected value of $f(\mathbf{r})$ when we intervene by setting the features in S to $\mathbf{x}_{i,S}$, denoted as $do(\mathbf{r}_S = \mathbf{x}_{i,S})$. This intervention is independent of any predefined joint probability distribution $p(\mathbf{x}, \mathbf{r})$. Therefore, the expression $\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S)$ represents the combined effect of the base expected value of $f(\mathbf{r})$ and the additional causal impact of the attribution $v^{(i)}(S)$.

⁵Note that $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$. Because both π and $\mathbf{p}_{\text{OT}}$ denote joint probability of $\mathbf{x}$ and $\mathbf{r}$, however, π makes the summation the minimum for the W_1 term across all possible joint distributions, whereas $\mathbf{p}_{\text{OT}}$ does not.

E PROOF OF THEOREM 5.1: COUNTERFACTUAL PROXIMITY

Theorem E.1 (Theorem 5.1 in the main text). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

Proof. Let $\mathbf{q}$ be the aligned reference matrix generated by $A_{\text{value}}(\mathbf{r}, \mathbf{p})$. For the elements where $c_{ik} = 1$ (modified elements), $z_{ik} = q_{ik}$. For elements where $c_{ik} = 0$, $z_{ik} = x_{ik}$. Therefore, we can write:

$$(z_{ik} - x_{ik})^2 = \begin{cases} (q_{ik} - x_{ik})^2, & \text{if } c_{ik} = 1; \\ 0, & \text{if } c_{ik} = 0. \end{cases} \quad (12)$$

The squared Frobenius norms with respect to $\mathbf{q}$ and $\mathbf{z}$ are computed as follows:

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2, \\ \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2. \end{aligned}$$

And,

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2 \\ &\stackrel{(i)}{=} \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{(i,k): c_{ik}=1} (q_{ik} - x_{ik})^2 \\ &= \sum_{(i,k): c_{ik}=0} (q_{ik} - x_{ik})^2 \geq 0, \end{aligned}$$

where the equality (i) holds because of equation (12).

Since the difference $\|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 \geq 0$, it follows that:

$$\|\mathbf{z} - \mathbf{x}\|_F^2 \leq \|\mathbf{q} - \mathbf{x}\|_F^2.$$

Taking square roots, we get:

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

Hence the conclusion. $\square$

Corollary E.2 (Corollary 5.2 in the main text). *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the factuals as the original counterfactual set reordered by σ .

Proof. Under the assumptions $n = m$ and $\varepsilon = 0$, the optimal transport plan $\mathbf{p}$ becomes a permutation matrix associated with a bijection σ . Consequently, the value assignment algorithm A_{value} (whether maximizing or averaging) assigns $\mathbf{q}_i = \mathbf{r}_{\sigma(i)}$ for every instance i . Substituting $\mathbf{q}$ with the permuted matrix $\mathbf{r}_\sigma$ into the general bound established in Theorem 5.1 (i.e., $\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F$) directly yields the result. $\square$

F ANALYSIS OF COMPUTATIONAL COMPLEXITY OF COLA

We perform analysis of the computational complexity of COLA as follows.

First, we analyze A_{Prob} . If the alignment between $\mathbf{x}$ and $\mathbf{r}$ is known a priori, then A_{Prob} just constructs the matrix $\mathbf{p}$ with the prior knowledge, which takes $\mathcal{O}(n \times m)$. Otherwise, we consider solving OT to obtain $\mathbf{p}$, which, by the Sinkhorn–Knopp algorithm, takes $\mathcal{O}(n \times m \times \log(1/\varepsilon))$.

Then we analyze A_{Shap} . For each subset, the model is evaluated on all n data points, leading to $\mathcal{O}(n)$ evaluations per subset. Incorporating baseline values from the reference data $\mathbf{r}$ involves replacing the values of certain features with their corresponding baseline values. This operation is $\mathcal{O}(m)$ because it requires accessing the baseline values from the reference table $\mathbf{r}$ for each of the d features. If we assume that the reference values can be precomputed and accessed in constant time, then the complexity of incorporating these values can be considered as $\mathcal{O}(d)$. The number of M_{Shap} subsets results in M_{Shap} model evaluations. Combining the above steps, the complexity of A_{Shap} is $\mathcal{O}(n \times d \times M_{\text{Shap}})$.

The normalization in line 4 of COLA takes $\mathcal{O}(n \times d)$.

To compare the complexities of the two algorithms, $A_{\text{Value}}^{\text{max}}$ and $A_{\text{Value}}^{\text{avg}}$, we analyze each algorithm step-by-step. For $A_{\text{Value}}^{\text{max}}$, for each row $\mathbf{x}_i$ in the data table, we need to (1) compute the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, which involves $\mathcal{O}(m)$ operations per row, (2) identify the row $\mathbf{r}_j$ in the reference data with the highest probability p_{ij} , which involves $\mathcal{O}(m)$ operations per row, and (3) assign $q_{ik} = r_{\tau(i),k}$ where $\tau(i) = \arg \max_j p_{ij}$, which involves $\mathcal{O}(d)$ operations per row. Since there are n rows in the data table, the total complexity for $A_{\text{Value}}^{\text{max}}$ is $\mathcal{O}(n \times (m + m + d)) = \mathcal{O}(n \times (2m + d)) = \mathcal{O}(n \times m + n \times d) = \mathcal{O}(n \times (m + d))$.

For $A_{\text{Value}}^{\text{avg}}$, for each row $\mathbf{x}_i$ in the data table, we need to 1) compute the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, which involves $\mathcal{O}(m)$ operations per row, 2) compute the sum $\sum_{j'=1}^m p_{ij'}$, which involves $\mathcal{O}(m)$ operations per row, and 3) calculate the weighted average $q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}$, which involves $\mathcal{O}(m \times d)$ operations per row. Since there are n rows in the data table, the total complexity for $A_{\text{Value}}^{\text{avg}}$ is $\mathcal{O}(n \times (m + m + m \times d)) = \mathcal{O}(n \times (2m + m \times d)) = \mathcal{O}(n \times (m + m \times d)) = \mathcal{O}(n \times m \times (1 + d)) = \mathcal{O}(n \times m \times d)$.

For lines 6–16, the entire complexity is straightforwardly $\mathcal{O}(n \times d) + \mathcal{O}(C) = \mathcal{O}(n \times d)$ due to the fact $C \leq n \times d$.

Therefore, the complexity of COLA using $A_{\text{Value}}^{\text{max}}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(n(m + d)) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nm) + \mathcal{O}(nd), \end{aligned}$$

and the complexity of COLA using $A_{\text{Value}}^{\text{avg}}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(nmd) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nmd). \end{aligned}$$

Hence the complexity of COLA with respect to n , m , d , and the regularization parameter ε of entropic OT is

$$\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N,$$

where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ if $A_{\text{Value}}^{\text{max}}$ is used and $N = \mathcal{O}(nmd)$ if $A_{\text{Value}}^{\text{avg}}$ is used.

G AN MILP FORMULATION OF EQUATION (1) WITH MEAND

In this section, we provide a global optimality benchmark for using a known alignment between factual and counterfactual in solving equation (1) with D being MeanD, namely

$$D(f(\mathbf{z}), \mathbf{y}^*) = \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{z}_i) - \bar{y}^* \right|,$$

with $\bar{y}^* = \frac{1}{m} \sum_{j=1}^m y_j^*$. Since COLA is used, we have $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$, and $\mathbf{z}$ stays closer to $\mathbf{x}$ than $\mathbf{r}$, hence equation (1b) is dropped. The formulation of equation (1) then becomes:

$$\min_{\mathbf{c}, \mathbf{z}} \left| \sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* \right|, \quad (13a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (13b)$$

$$z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik}), \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (13c)$$

Note that the original constraints equation (1d) and equation (1e) merge to be equation (13c), because CF- p_{Ect} is imposed to be used. That is, for any $\mathbf{x}_i$, there is an exact $\mathbf{r}_j$ serves as its reference in A_{Shap} and A_{Value} , such that x_{ik} ($k = 1, 2, \dots, d$) either stays unchanged or can be changed to r_{jk} . Therefore $z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik})$ of which the value depends on the binary variable c_{ik} .

Due to the known alignment between any $\mathbf{x}_i$ and its corresponding $\mathbf{r}_j$, $\mathbf{q}$ is determined (also, both $A_{\text{Values}}^{\text{max}}$ and $A_{\text{Values}}^{\text{avg}}$ return the same $\mathbf{q}$). For any data point i and any feature set $S \subseteq \mathcal{F}$, let $\mathbf{z}_{iS}$ denotes the solution $\mathbf{z}_i$ where we have all features $k \in S$ changed to q_{ik} , and the other features $h \in \mathcal{F} \setminus S$ stays x_{ih} . Hence the set of $\mathbf{z}_{iS}$ ($S \subseteq \mathcal{F}$) composes the domain of all possible values of $\mathbf{z}$. Define a corresponding scaler variable for any $\mathbf{z}_{iS}$:

$$g_{iS} = f(\mathbf{z}_{iS}) - \bar{y}^*.$$

Then, for any $\mathbf{z}_i$ in equation (13), the value of the term $\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^*$ can be represented by a binary variable a_{iS} together with the scaler g_{iS} , namely,

$$\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* = \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS}.$$

The optimization problem equation (13) is hence reformulated as a mixed integer programming below.

$$\min_{\mathbf{a}, \eta} \quad \eta, \quad (14a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \leq \eta, \quad (14b)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \geq -\eta, \quad (14c)$$

$$\sum_{S \subseteq \mathcal{F}} a_{iS} = 1, \quad i = 1, \dots, n, \quad (14d)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} |S| a_{iS} \leq C. \quad (14e)$$

Minimizing η under constraints equation (14b) and equation (14c) is equivalent to minimizing the objective in equation (13). Constraints equation (14d) ensure that each data point i is assigned exactly one feature-modification plan a_{iS} for some $S \subseteq \mathcal{F}$, while constraint equation (14e) corresponds to equation (13b). Solving equation (14) therefore provides the theoretical optimum for COLA when factual-counterfactual alignment is known, as illustrated in Figure 4 in Section 6.

H EXTENDED NUMERICAL RESULTS

Across Figures 5–8, CF- p_{Uni} generally improves upon RB- p_{Uni} . Likewise, comparing RB- p_{OT} with CF- p_{OT} , which also differ only in A_{Shap} , shows that the latter consistently dominates the former. These patterns indicate that RB-SHAP is not suitable for FA in CE.

We analyze how alternative Shapley methods affect FA, corresponding to lines 3–4 of COLA. The methods separate into comparisons that isolate the attribution mechanism: RB- p_{Uni} and CF- p_{Uni} differ

only in A_{Shap} , and Figures 5–8 show that $\text{CF-}p_{\text{Uni}}$ generally performs better; similarly, $\text{RB-}p_{\text{OT}}$ and $\text{CF-}p_{\text{OT}}$ also differ only in A_{Shap} , and the latter consistently outperforms the former. Hence *RB-SHAP is not suitable for FA in CE*.

Beyond FA, line 5 constitutes an equally important component of COLA: the joint probability $p(\mathbf{x}, \mathbf{r})$ is used to construct $\mathbf{q}$, thereby specifying the directions in which factual features should be changed to approach the target model outcome. In Figures 5–8, $\text{CF-}p_{\text{OT}}$ consistently outperforms the alternatives across the experiments. Although $\text{CF-}p_{\text{Uni}}$, $\text{CF-}p_{\text{Rnd}}$, and $\text{CF-}p_{\text{OT}}$ all provide candidate joint probabilities $\mathbf{p}$ when exact factual–counterfactual alignment is unavailable, their performance differs markedly, showing that knowledge of the CE and its marginal distribution alone is insufficient for p_{OT} .

OT is especially useful when alignment information between factual and counterfactual instances is absent or unreliable. Even when a CE algorithm explicitly pairs each factual instance with a corresponding counterfactual, there is no guarantee that this known pairing is performance-optimal, as supported by Figure 4 in Section 6.

Importantly, p_{OT} need not coincide with the true data-generating joint distribution of $\mathbf{x}$ and $\mathbf{r}$; rather, it should provide a useful coupling that enables COLA to jointly process $\mathbf{x}$ and $\mathbf{r}$ for both FA and CE. The QDA column in Figure 5 further illustrates the stability of OT-based methods, whereas other approaches diverge substantially from the target. Nevertheless, COLA is *not limited to using OT* as A_{Prob} . As Figure 4 indicates, even the best known $\mathbf{p}$ may retain a non-negligible gap to global optimality, making the design of improved A_{Prob} algorithms a natural direction for future investigation.

I EXPERIMENTS REPRODUCIBILITY

The experiments were conducted on a high-performance computing (HPC) cluster, using four nodes in parallel for the four datasets; each node was equipped with two Intel Xeon Processor 2660v3 CPUs (10 cores, 2.60GHz) and 128 GB of memory. Runtime was approximately 5–10 hours per node, depending on dataset size. The experiments can also be reproduced on a laptop, although this generally requires substantially more computation time than the HPC configuration.

For all four datasets, numerical features are standardized, while categorical features are encoded using either label encoding or one-hot encoding. In practice, these two categorical-encoding choices did not produce a notable difference in COLA’s performance. The train–test split follows a 7 : 3 ratio.

The optimality baseline in Figure 4 is solved with Gurobi 11.0.2 ([gur](#)); reproducing this baseline therefore requires a Gurobi license, although open-source operations-research libraries such as Google-OR Tools ([goo](#)) may be used as alternatives. We emphasize that solving the MILP in Appendix G is computationally expensive and is therefore practical only for small-scale datasets such as German Credit. Computing the optimality baseline for larger datasets would require reducing the number of included features.

Model hyperparameters are specified as follows. Bagging, GP, RBF, RndForest, AdaBoost, GradBoost, and QDA are implemented with scikit-learn ([skl](#)) using default hyperparameters. DNN, SVM, RBF, and LR are implemented in PyTorch ([Paszke et al., 2019](#)); the DNN has three layers, and the SVM uses a linear kernel. XGBoost ([Chen & Guestrin, 2016](#)) and LightGBM ([Ke et al., 2017](#)) are used through their scikit-learn interfaces, again with default hyperparameters.

THE USE OF LARGE LANGUAGE MODELS (LLMs)

Large language models (LLMs) were used only as general-purpose assistance tools, and their role was restricted to grammar, wording, and light copy-editing of author-written text. They did not contribute to research ideation, modeling choices, experimental design, or results. All algorithms, proofs, datasets, and analyses were created and verified by the authors, and any LLM-suggested phrasing was reviewed and edited before inclusion. The authors take full responsibility for the manuscript content; LLMs are not authors or contributors. This disclosure is provided in accordance with the ICLR policy on LLM usage.

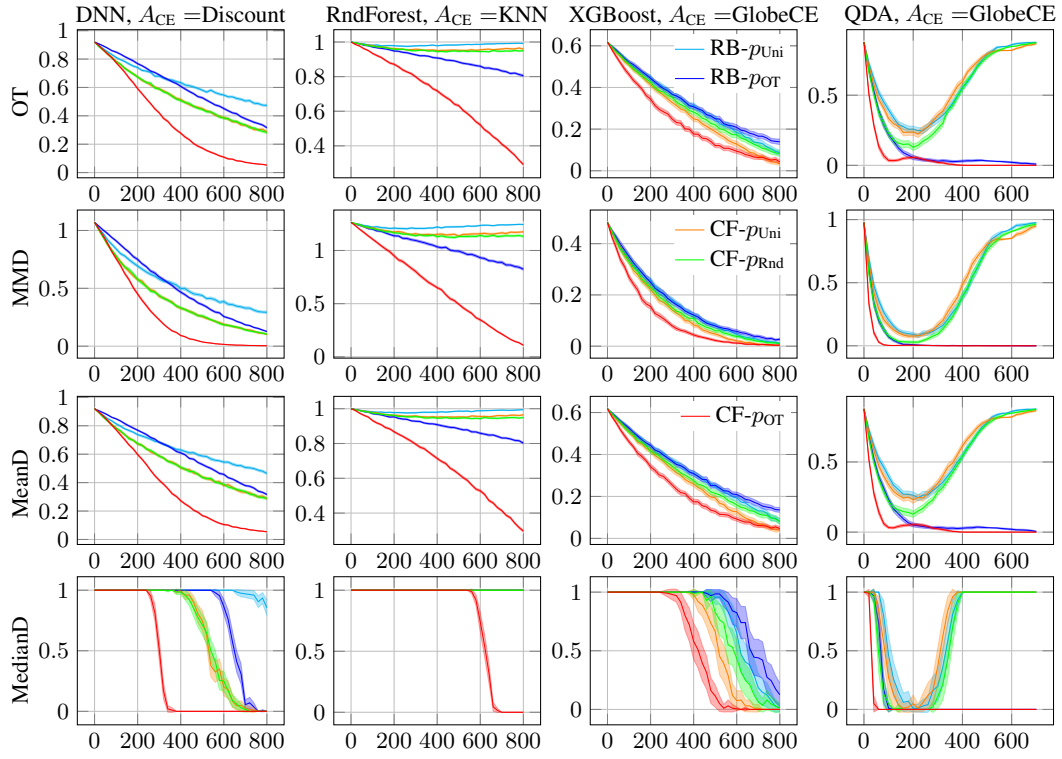


Figure 5: [HELOC] $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of allowed actions C , averaged over 100 runs. Shaded regions denote 99.9% confidence intervals, legends apply to all panels, and $A_{\text{Value}}^{\text{avg}}$ is used.

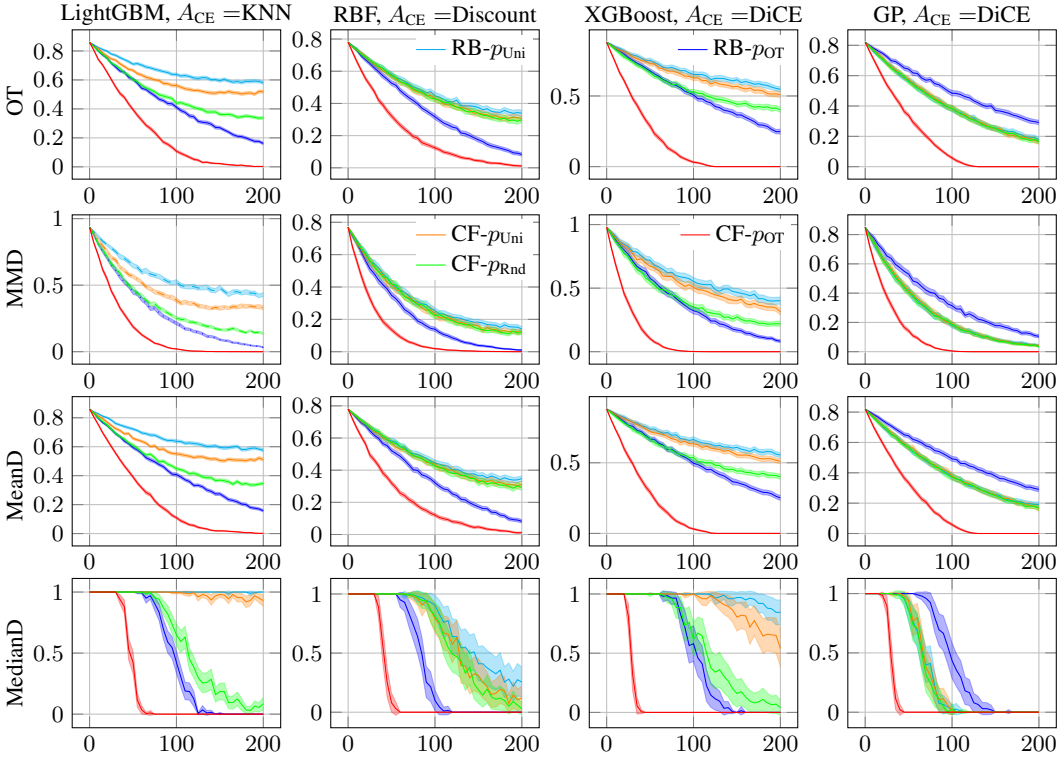


Figure 6: [German Credit] $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of allowed actions C , averaged over 100 runs. Shaded regions denote 99.9% confidence intervals, legends apply to all panels, and $A_{\text{Value}}^{\text{avg}}$ is used.

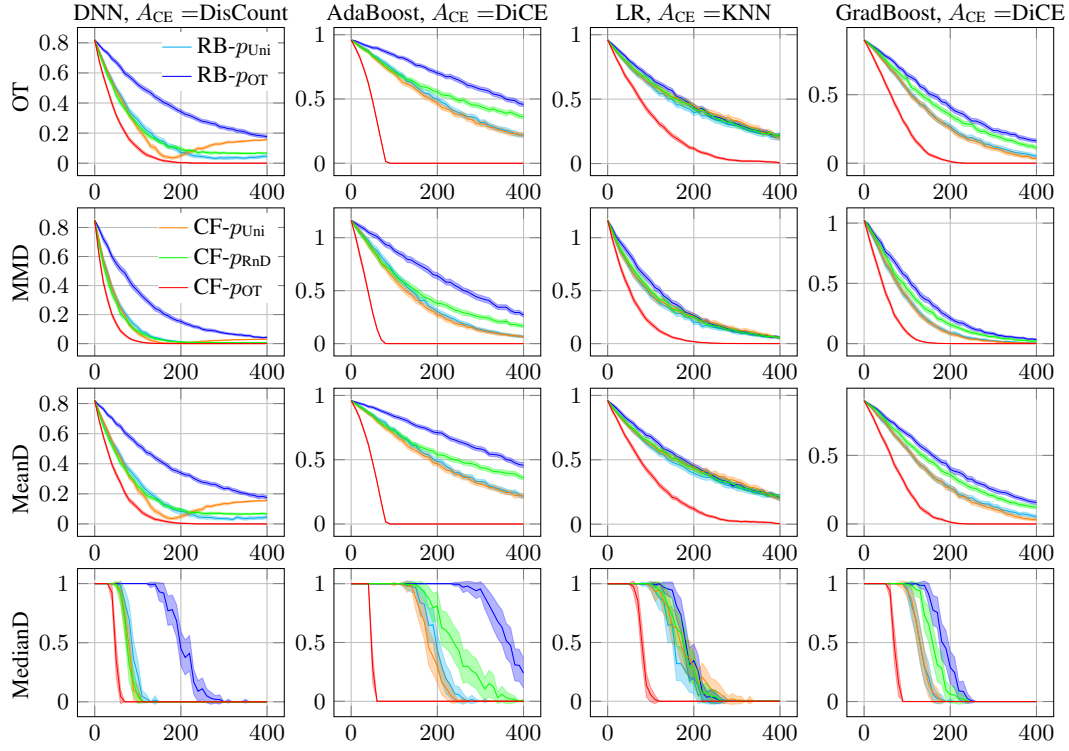


Figure 7: [Hotel Bookings] $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of allowed actions C , averaged over 100 runs. Shaded regions denote 99.9% confidence intervals, legends apply to all panels, and $A_{\text{Value}}^{\max}$ is used.

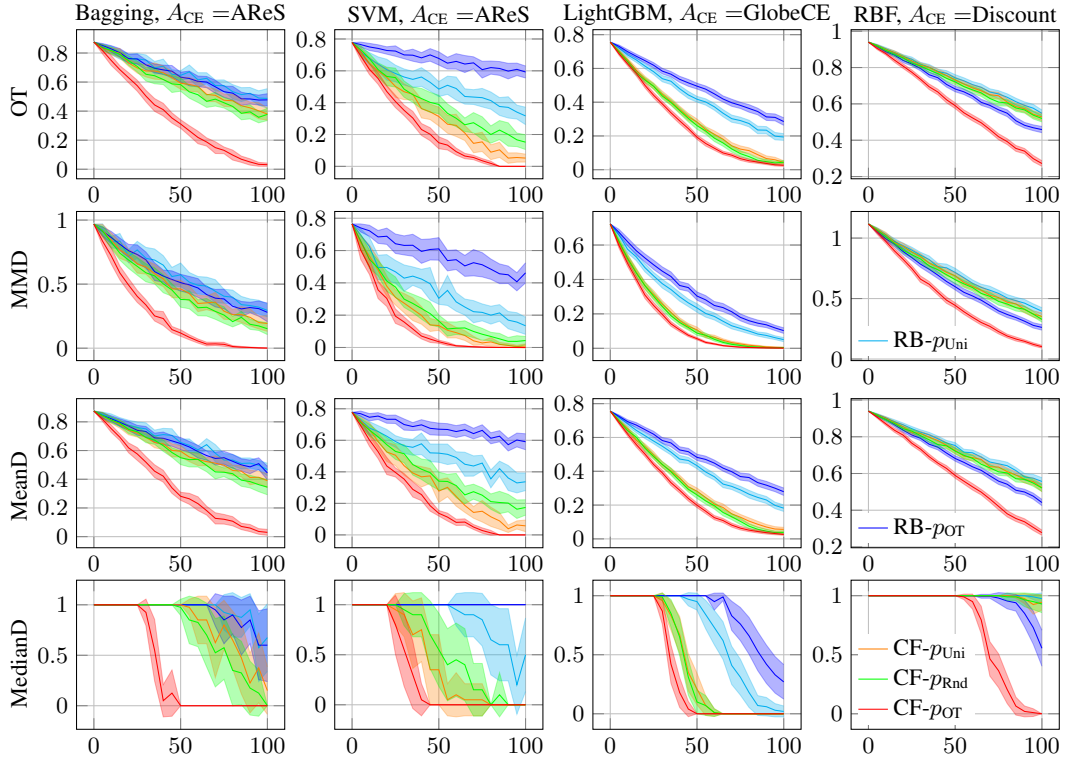


Figure 8: [COMPAS] $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of allowed actions C , averaged over 100 runs. Shaded regions denote 99.9% confidence intervals, legends apply to all panels, and $A_{\text{Value}}^{\max}$ is used.